



Silver

△ **English sterling-silver** christening mug of Edward VIII, in Art-Deco style

Both gold and silver have been used as currency for thousands of years, and although gold has become synonymous with wealth, silver is now increasing in value because of its scarcity. Since pure silver is easily damaged, jewellers work with sterling silver, which has copper added for strength. Silver in folklore is often related to the moon, a link made much of by silversmiths including legendary Danish designer Georg Jensen, who worked in the early 20th century. Since then, silver has grown in demand for use in both jewellery and industry.

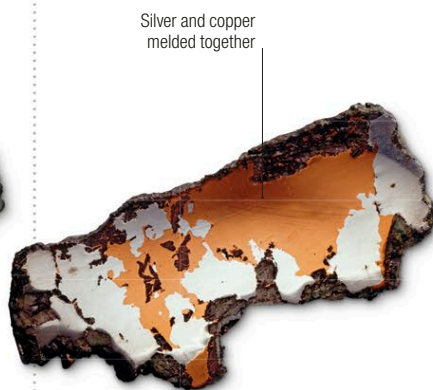
Specification

Chemical name Silver | **Formula** Ag | **Colour** Silver
Structure Cubic | **Hardness** 3.25 | **SG** 10.1–11.1
Streak Silver-white | **Locations** Mexico is the greatest single producer; also Peru, USA, Canada, Norway, Australia, Russia, Kazakhstan

Rough



Wiry silver | Most silver in the ground is extracted from ores but, in its native state, silver can also appear as a coarse mass of tendrils. Here, it is growing in quartz.



Polished silver and copper slice | Native copper and native silver can sometimes form together in a single specimen, commonly known in the USA as a “half-breed”.



Tarnished silver | The surface of silver is susceptible to tarnishing, which appears as a coating over the surface when exposed to oxygen or hydrogen sulphide, as here.

Settings

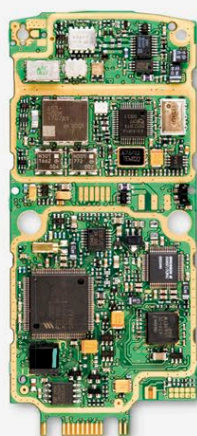


Moonlight brooch | Georg Jensen likened silver to the glow of the moon, and here he combines sterling silver with moonstones in an Art Nouveau design.

Silver in industry

Demand outstrips supply

Silver is naturally more abundant than gold in the Earth's crust, but over the past 20 years, stocks have dwindled dramatically due to the fact that it has become highly sought after in manufacturing for its exceptional ability to conduct both heat and electricity. Silver is a key component in photovoltaic panels for solar power, a growing sector that will only increase the demand for silver as an industrial commodity. It is also used in the production of almost all electronic devices, from circuit boards and TV screens to mobile phone batteries and computer chips. A mobile circuit board typically contains about 300mg of silver.



Mobile phone circuit board | Silver is used in circuits such as this.



Silver dendrite | Some silver crystals grow in a dendritic (branching) formation, which has formed a naturally tree-like structure in this specimen.



Silver fish | This 1940s sterling silver bracelet by Margot de Taxco is in the shape of a koi. De Taxco lived and worked in Mexico, the world's top silver-producing country.